

Setup Guide: Light Seeker

Time Allocation: 90 Minutes

Challenge Overview

Using light sensors, the robot must locate a flashlight source placed at the far end of a darkened room.

Environment & Arena Setup

- **Room Preparation:** Select a classroom or lab space that can be made completely dark by closing blinds, turning off overhead lights, and blocking any significant light gaps under doors.
- **Target Placement:** Place a bright, steady-beam flashlight (or a smartphone flashlight on a stable stand) at one end of the testing area, pointing horizontally across the floor toward the starting zone.
- **Course Boundaries:** Use tape or barriers to create a clear lane approximately 6 to 8 feet long and 3 to 4 feet wide to guide the general path and prevent robots from wandering into classroom furniture.
- **Safety Check:** Ensure the floor is clear of loose cables and obstacles that could trip students or snag robot wheels during testing.